Tinashe Shoniwa

Mechanical Engineering Student · BSc Candidate, Hochschule Rhein-Waal

Kleve, Germany · +49 176 848 293 97 · shoniwatinashe7@gmail.com · LinkedIn · Portfolio

PROFILE

Mechanical Engineering undergraduate combining hands-on machining and assembly experience with strong CAD and simulation skills. SolidWorks CSWA-certified, proficient in ANSYS Mechanical (linear-elastic and elastic-ideally-plastic) and confident applying the FKM guideline for static and fatigue strength assessment. Comfortable taking a design from free-body diagram through CAD, FEA, and manufacturing documentation. Seeking a mechanical engineering internship in design, simulation, or production engineering.

EDUCATION

Hochschule Rhein-Waal — BSc Mechanical Engineering, Focus Field: Design *Kleve, Germany | Oct 2022 – present*

- Relevant coursework: Finite Element Methods, Modelling & Simulation, Engineering Design, Material Testing, Statics & Strength of Materials, Dynamics, Control Systems, Numerical Methods, Programming.

Global English School — Higher Secondary (Mathematics, Physics, Chemistry) *Abu Dhabi, UAE | Sep 2016 – Jun 2022*

TECHNICAL SKILLS

- **CAD & Modelling:** SolidWorks (CSWA-certified; Sheet Metal, Surfacing, ScanTo3D, Cavity Tool), Siemens NX, AutoCAD
- **Simulation & Analysis:** ANSYS Mechanical / Static Structural (linear-elastic and elastic-ideally-plastic), FKM guideline (static & fatigue, local stress)
- **Manufacturing:** CNC lathe & drilling, GD&T, dimensional inspection (vernier callipers, micrometers), bearing installation, hydraulic press, surface finishing, technical-drawing interpretation (Prüfplan)
- **Reverse Engineering & Metrology:** CREAFORM HandySCAN 300, VXelements, STL → CAD surface reconstruction, mould cavity/core design
- **Quality & Production:** Process FMEA (S×O×D, RPN scoring), takt-time & capacity calculation, factory-layout planning, batch sizing
- **Programming & Tools:** MATLAB & Simulink, Python, LaTeX, Microsoft Office, Adobe Photoshop

KEY PROJECTS

FEM & FKM Strength Assessment of a Racing-Car Seat Shell — *Solo project, Finite Element Methods* *Feb 2026*

- Verified a 1.5 mm Al7075-T6 racing seat shell under two load cases (9g vertical circuit-driving load and 20g rear-crash with a 120 kg driver) in ANSYS Mechanical 2025 R2, running both linear-elastic and elastic-ideally-plastic models.
- Drove mesh convergence through iterative edge sizing, biasing, and inflation until quality exceeded 70% at all hotspots, then validated stress stability across coarse/fine meshes.
- Applied the **FKM guideline (Chapters 3–4)** for static and fatigue assessment with local stresses; reported degree of utilization of 0.71 (Load Case 1, static pass) and demonstrated the design fails the 10^8-cycle fatigue requirement and the 20g rear-crash static check, with concrete redesign recommendations.

"Il Bombo" Production-Site Planning — Piaggio Scooter Series — *Quality & Production Management* *WS 2025/26*

- Designed a full production facility for **120,000 units/year**, calculating takt time (168.75 s/unit on a 3-shift, 22.5 h/day model) and sizing machinery and workforce for metal stamping, robot spot/MAG welding, injection moulding, painting, and assembly.
- Benchmarked 1-robot vs. 2-robot welding cells (cycle times 9.92 vs. 7.3 min) against per-cell investment (€1.26 M vs. €1.38 M) to select an optimum configuration; computed total capital investment across all stages.
- Authored a full **Process FMEA** covering raw-material delivery, coil cutting, and forming; quantified Risk Priority Numbers and verified RPN reductions after corrective actions (e.g., supplier audits, batch barcode scanning, inline edge-tracking).

Bevel Gearbox — Concept to CAD Assembly — *Team of 2, Engineering Design* *Apr – Jun 2025*

- Designed a complete bevel gearbox from a selected gear pairing (Maedler 36144800, 16/24 teeth, 33.69° pitch angle); calculated tangential, radial, and axial gear forces from input torque using a 20° normal pressure angle.
- Sized C45 driving shaft (Ø15 mm) and 42CrMo4 driven shaft (Ø20–24 mm); verified bending, torsional, and von Mises equivalent stresses against material allowable (245 MPa and 500 MPa) with an application factor of 1.5.
- Selected real SKF/NTN tapered-roller and angular-contact bearings from manufacturer catalogues, designed parallel-key connections, housing, and seals, and produced full SolidWorks shaft and assembly drawings to drawing-standard specifications.

Reverse Engineering of a Heat-Pump Tumble-Dryer Fascia — *Virtual Product Design* *Jan 2025*

- Captured an OEM injection-moulded control panel (part 0107) using a **CREAFORM HandySCAN 300** (0.04 mm accuracy, 205,000 measurements/s) and processed the resulting mesh in VXelements.
- Imported the STL into SolidWorks via ScanTo3D, reconstructed curved surfaces and bosses despite limited mesh resolution, and designed a manufacturing mould (cavity + core) using SolidWorks' Cavity Tool.u

Scissor-Lift Table — Statics & Stress Analysis — *Solo project* *Dec 2024*

- Calculated reaction forces, internal member forces, and pin shear stresses for a 5 kN structural-steel scissor-lift across three working heights using free-body diagrams and equilibrium equations.
- Sized the leg-connector pin (τ = 8.09 MPa) and verified leg bending stresses with the flexure formula; modelled the full assembly in SolidWorks.

EXPERIENCE

Rotterberg Maschinenbau — *Mechanical Engineering Intern* *Kleve, Germany | Aug – Sep 2023 (4 wks)*

- Machined precision shafts to tight tolerances from customer *Prüfpläne* (lathe turning, drilling counterbore holes, sawing) in a series-production environment.
- Performed dimensional inspection of every batch with vernier callipers; identified a blunted saw blade and escalated to the supervisor, restoring throughput without scrap.
- Installed bearings into a batch of 35 assembled rods with retention bands and finished surfaces to customer roughness specifications; passed 100% of inspections on first review.

Civils360 / Bulawayo City Council — *Mechanical Engineering Intern* *Bulawayo, Zimbabwe | Jul – Aug 2023 (4 wks)*

- **Technical procurement:** ran the supplier RFQ cycle for engineering materials (drafting enquiry emails, comparing quotations, raising orders, supervising delivery and barcode-tracked stock intake) and managed defect returns with reimbursement.
- **Equipment servicing:** identified, measured, replaced, and lubricated bearings using micrometers, torque wrench, and bearing puller/press; logged work in a CMMS-style maintenance system with scheduled follow-ups for water-plant motors, blowers, and submersible pumps.
- **Machining:** threaded DN 900 gate-valve shafts on a manual lathe (with inch-to-mm conversion) and drilled metal plates under PPE protocols.

CERTIFICATIONS & LANGUAGES

Certifications	**Languages**
SolidWorks CSWA *(Certified SolidWorks Associate)*	English (Native)
IELTS Academic — Band 7 *(CEFR C1, English)*	Shona (Native) · Ndebele (Native)
MATLAB Simulink	Zulu (Native)
MATLAB ONRAMP	German — B1 (actively improving)